

Week 10: Lambda terms and evaluation

Functional Programming in Lean

Learning goals

- ▶ Inductive lambda-term syntax
- ▶ De Bruijn indices
- ▶ Call-by-value evaluation with fuel

Core concepts

- ▶ Variable, abstraction, application
- ▶ Index zero: nearest binder
- ▶ Closure: body and defining environment

Lean example

```
inductive Term where  
  | var (index : Nat)  
  | lam (body : Term)  
  | app (function argument : Term)
```

Call-by-value evaluator

```
def evalCBV :  
    Nat → List Value → Term → Option Value
```

- ▶ Closure: body and environment
- ▶ Fuel: evaluation bound

Lean details

- ▶ `wellScoped`: bound-variable check
- ▶ `evalCBV`: environment-based evaluator
- ▶ Call-by-value versus call-by-name traces

Live coding

- ▶ Identity term
- ▶ `wellScoped`
- ▶ `evalCBV` on self-application

Check

```
wellScoped 0 (.var 0)
```

Exercise

- ▶ De Bruijn translations
- ▶ Evaluation trace